Promising Less to Be Wrong Less Often: Four Renunciation Mechanisms as Alternative Actions in Crop Mapping

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Abstract. An automatic crop map that is unsure of a parcel can do more than answer anyway. It can narrow the catalogue it promises, decline to answer, hand back a small set of candidate labels, or fall back to a coarser agronomic category. These are four different ways of promising less, and the literature treats them as separate lines of work: the error–reject trade-off, set-valued prediction, catalogue granularity, and label hierarchies each have their own formulations, their own optima, and their own evaluations. Within the search reported in this paper, we found no comparison that reads them as alternative actions in one decision problem, scored under one declared loss, with an account of who bears that loss. This paper sets up that comparison. We state the estimand and the population before computing anything, we hold the operating point of every mechanism to training and validation data only, and we treat the price of each outcome as a quantity to be elicited from the people who act on the map rather than assumed from the size of the answer. We report what such a comparison requires and what it forbids, and we are explicit about which of our claims are conditional on an elicitation that has not yet been carried out. The mechanisms, the estimand, and the accounting of distribution across classes are described here; the empirical verdict is deliberately withheld until its governing loss exists.

Keywords: Crop mapping · Selective classification · Set-valued prediction · Decision theory · Satellite image time series

1 Introduction

A parcel-level crop map is a promise. For every field in the scene it names a crop, and somebody downstream acts on that name: an inspector schedules a visit, an agronomist plans an input, an agency settles a subsidy. When the classifier is unsure, answering anyway transfers the uncertainty to that person without telling them.

There is more than one way to say less. The map can *shrink the catalogue* it is willing to promise and stop offering the classes it cannot separate; it can *abstain* on a parcel and hand it to a human; it can return a *set of candidate labels* and let the reader pick; or it can *fall back* to a coarser agronomic category that it can defend. Each of these buys accuracy with a different currency, and each one moves the cost somewhere else.

Promising less to be wrong less often is not a new idea. The error–reject trade-off is due to Chow [3], and Ha later generalized it to selecting a *subset* of classes—together with the optimality criterion that governs it [10]. Recent work gives efficient algorithms for the subset of maximum expected utility [13] and a systematization of the available formulations [4]. In crop mapping specifically, catalogue size has already been measured as a design variable [7], and the closest work to ours rejects pixels by uncertainty on the same benchmark and on three architectures also represented in our original candidate bank, making the trade-off between precision and recall explicit [15].

What we did not find is a comparison of these mechanisms *against each other*. Each line of work optimizes its own device and reports its own curve. Within the search scope described in Section 2, we found no work that puts catalogue shrinking, abstention, set-valued answers, and taxonomic back-off in one decision problem, prices them with a single declared loss, and asks which classes end up paying for the improvement.

Contribution. We state the comparison and the conditions under which it can be believed. Concretely: (i) we cast the four mechanisms as alternative *actions* of the same decision problem, so that a single loss can score all of them; (ii) we fix the estimand, the population, and the unit of analysis *before* computing anything, and we require the operating point of every mechanism to come from training and validation data alone; (iii) we make the distribution of the loss across classes a reported quantity rather than a side effect, because a mechanism that improves the average by silently dropping the rare classes has not improved the map for the people who care about those classes.

What this paper does not yet claim. The loss that scores the four actions is meant to be elicited from practitioners who act on crop maps, not assumed from the size of the answer. That elicitation is specified and has not been carried out. Until it exists, this paper does not report which mechanism is preferable, nor at whose expense: the framework, the estimand, and the accounting are presented here, and the empirical verdict is deliberately withheld. We consider stating a preference under an assumed price to be the failure mode this work exists to avoid.

2 Related Work

We organize this section by *limitation* rather than by application: for each line of work we state what it establishes and where it stops, because the stopping point is what defines the work that is left.

The renunciation framework already exists. The error–reject trade-off and its optimal rule are due to Chow [3]; that formulation is the degenerate case in which the only alternative to answering is silence. Ha generalized it to class-selective rejection, showing that selecting a subset of classes is the optimal compromise between error and the mean number of classes returned [10]; the treatment is decision-theoretic and assumes the posteriors are known. Mortier et al. give algorithms for the Bayes-optimal subset of maximum expected utility, with a utility that is a function of set size [13], and Chzhen et al. unify the existing set-valued formulations and their trade-offs [4], at infinite sample and under a plug-in principle. The framework is therefore not our contribution, and we say so on the first page. What these treatments share is that the price of an answer is a function of *how large* the answer is. Neither the party who bears the loss nor its distribution appears in them.

Learned abstention is developed, and out of our scope by choice. Deep selective classification supplies risk guarantees [5], an architecture with an integrated reject head [6], a wagering-style loss [12], and an AUC-based selection criterion [14]. All of them abstain *per instance*; none emits a set or falls back to a coarser class. We do not reimplement them and we do not benchmark against them: our question is how abstention compares with the other three actions under one loss, not whether a better abstention rule exists.

Renunciation has reached Earth observation, and not through agriculture. The closest work to ours rejects pixels by an uncertainty threshold on PASTIS, using three architectures represented in our original candidate bank, and makes the trade-off explicit: not answering lowers recall and can raise precision when the rejected pixel would have been misclassified [15]. That is the risk–coverage compromise, performed and acknowledged. What it does not do is place the decision in the selective-classification frame, compare alternative actions, or model who absorbs the loss. Elsewhere in Earth observation, SHRUG-FM reports abstention with interpretable thresholds [9], on burn scars, floods, and landslides rather than on any agricultural task. Rejection methods have been applied to vegetation mapping [8], on airborne hyperspectral data rather than per-parcel satellite time series. Open-set crop recognition rejects the *unknown* through outlier exposure [2], which is a different question from distributing a fixed budget of promise among known classes.

Catalogue size is already a measured design variable. Ghassemi et al. treat the number of classes as a design variable and measure it: starting from 52 LUCAS classes, a 26-class scheme balances accuracy against detail [7]. This is the direct precedent for reading catalogue shrinking as an operating point, and it forecloses any claim that we are the first to do so. There the setting is land cover with Random Forest; here it is one mechanism among four, scored under a declared loss.

Label hierarchies exist, with another purpose. Multi-scale label hierarchies improve crop mapping from satellite image time series [16], and hierarchical fusion combines satellite, rotational, and contextual signals [1]. In both, the hierarchy serves *training* or information fusion. Using it at inference as a way of answering at a coarser granularity—taxonomic back-off as an action—does not follow from either, and we treat it as a mechanism to be evaluated rather than as established practice.

Who pays is documented, outside our domain. Selective classification can *magnify* disparities between groups: abstaining more can make outcomes worse for some of them [11]. The evidence is on demographic groups in vision and language, not on agronomic classes or parcels. We take it as the reason to report distribution across classes rather than as evidence about crops, and we note that the result runs opposite to the intuition that declining to answer is a neutral act.

Novelty, stated with its universe. We found no work that compares catalogue shrinking, per-parcel abstention, set-valued prediction, and taxonomic back-off *as alternative actions of the same decision problem*, under a loss declared by action, outcome, and affected party, with an accounting of how that loss is distributed across classes. This statement is bounded by our search: arXiv, OpenAlex, and Semantic Scholar, 61 queries plus 6 manual ones, covering 2019–2026 together with the classics retrieved by name, in English only. That procedure indexes remote-sensing proceedings without DOIs poorly, and it is written in the vocabulary of selective classification—a limitation we observed in practice, since the closest work to ours [15] uses none of those terms and was found by another route.

3 Problem Statement and Estimand

3.1 Four Ways to Promise Less

Let $\mathcal{Y}$ be the class catalogue, let x_i be the observations for parcel i, and let $y_i \in \mathcal{Y}$ be its reference label. A conventional classifier commits to a map $x_i \mapsto \hat{y}_i \in \mathcal{Y}$: it must name exactly one class for every parcel, whatever its posterior looks like. The mechanisms compared here all relax that commitment in the same formal way. Each replaces the point prediction with a subset of the catalogue,

$$C_m(x_i) \subseteq \mathcal{Y}, \qquad C_m(x_i) = \emptyset \text{ permitted}, \tag{1}$$

and they differ only in which subsets they may emit, and in what fixes the choice.

Catalogue shrinking. A subcatalogue $\mathcal{S} \subset \mathcal{Y}$ is fixed before any parcel is seen. A parcel whose unrestricted posterior maximizer lies inside $\mathcal{S}$ receives the restricted maximizer; a parcel whose maximizer lies outside receives nothing. The rule reads the posterior and never the reference label—keying delivery on the truth would make the mechanism unimplementable at inference time, and an earlier implementation in this line of work did exactly that. The decision is per class, and it is taken once.

Per-parcel abstention. The mechanism emits a singleton or the empty set, according to whether a confidence score clears an operating point. The decision is per parcel, and it is taken as many times as there are parcels.

Set-valued prediction. The mechanism emits a non-empty subset whose size varies with the difficulty of the parcel, calibrated so that a stated guarantee holds over the population. Every parcel receives an answer; what varies is how much the answer commits to.

Taxonomic back-off. The mechanism emits the set of leaves under a node of a class hierarchy. Formally this is a subset like any other; what distinguishes it is that the admissible subsets are those an agronomist can name, so the answer degrades to a coarser but still communicable category instead of to a list.

Two degenerate elements of the same space bound it: full-catalogue singleton prediction, which renounces nothing, and universal non-delivery, for which $C \equiv \emptyset$. Neither is one of the mechanisms, because neither chooses what to give up. They are not on the same footing, however, and Section 4 keeps them apart: the unmodified singleton is evaluated, as the reference against which every renunciation is priced, whereas universal non-delivery only fixes the price of silence and is never run. Writing all of these in one action space is this paper's first move, and it is what makes the question askable at all: as long as legend design, thresholding, conformal calibration, and hierarchical prediction are treated as separate techniques, there is no object on which to ask which of them one should prefer.

3.2 The Estimand

Fix a dataset d with its region, campaign, split, and predictor. For an affected party a and a mechanism m operating at τ_m, the quantity this paper estimates is the mean loss over the held-out block,

$$R_{d,a,m} \;=\; \frac{1}{N_d} \sum_{i=1}^{N_d} L_a\big(y_i,\, C_{m,\tau_m}(x_i)\big), \tag{2}$$

where N_d counts *all* eligible parcels of that block. Three properties of (2) carry most of this paper's argument, and each is a commitment rather than a convenience.

The loss is not defined here. L_a is an elicited object, not a modeling choice: it prices a wrong label, a missing answer, an ambiguous set, and a coarsened category against one another, and those prices differ by who is reading the map. We deliberately do not supply it. Expected set cardinality, the quantity most readily available, is not a candidate: it prices two sets of equal size identically however useful each is, and it prices the empty set at zero, which is precisely the accounting error that makes abstention look free. Until L_a exists, the comparison licensed by (2) is defined but not computed, and this paper reports the framework without it. Assuming a loss and then reporting which mechanism wins under it would hide the assumption and present its consequence as a finding.

The population includes the parcels that receive nothing. N_d ranges over every eligible parcel in the block, including those the mechanism declines to answer and those whose class the shrunken catalogue no longer contains. Restricting the denominator to delivered parcels would price renunciation at zero and would score each mechanism on a population of its own choosing, so that a mechanism could improve its reported quality by answering less.

The operating point is fixed before the held-out block is touched. τ_m is derived entirely from training and validation. The held-out block estimates the realized outcome and nothing else. Re-equalizing loss, coverage, or delivery rate on the held-out block is prohibited, including as a post-hoc alignment intended to make mechanisms comparable: it is the leak that two audits of an earlier version of this work found, first in the threshold and then in its target rate.

3.3 Scope, Dependence, and What Is Not Claimed

Inference is *conditional on each dataset*, with its region, campaign, split, and predictor. The observational unit is the parcel. Parcels are not independent: they arrive in image patches that share acquisition, geography, and annotation, so the dependence cluster is the patch. The inferential bootstrap therefore resamples whole patches within spatial blocks and never parcels one by one. A separate paired interval over spatial-block means is reported beside it to expose between-block variation; because those folds share training data, it is a sensitivity view rather than evidence from independent territories. Below three unique paired units for either view, that view is reported descriptively — without interval, without p-value, and without multiplicity correction — because at that size the interval would describe the resampling scheme and not the data.

No quantity is pooled across datasets. Each dataset is reported on its own, and the second one serves as a replication attempt rather than as an additional sample of the first. The number of spatial blocks is a sensitivity parameter for the spatial structure of the estimate, not a count of replicates and not a lever on statistical power: partitioning the same territory more finely makes the resulting folds share more of their training data, so it produces the same evidence counted more often.

What this design gives up is stated here rather than in a limitations paragraph, because it constrains every claim that follows: **we do not claim transport**. Nothing reported for one region or campaign is asserted to hold in another. A conditional estimand is what the available partitions support; a marginal one would require a sample of regions that this work does not have, and the two were mixed in the design of an earlier version until an audit separated them.

The intended deployment context is Mexico, but our bounded search identified no public Mexican benchmark with parcel-level crop labels. The datasets available to this study therefore do not validate the deployment geography. This is a limitation of the evidence, not proof that such data do not exist, and no result below is generalized from the evaluated regions to Mexico.

4 Four Mechanisms as Alternative Actions

The comparison contains four mechanisms but six experimental arms. This distinction matters. There is one unmodified singleton baseline; catalogue shrinking has two arms, because the retained catalogue is ranked once by validation F-score and once by training support; and abstention, set-valued prediction, and taxonomic back-off contribute one arm each. Universal non-delivery is a boundary action used to establish the price of silence, not a seventh experimental arm. Thus the count is

$$\text{baseline} + \text{two catalogue rules} + \text{three other mechanisms} = \text{six arms},$$

while the scientific question remains a comparison of four ways to renounce commitment. This also prevents a bookkeeping ambiguity in Section 3.1: the two boundary actions define the space, but they are not the two extra empirical arms.

4.1 A Common Output Type

Every arm returns the Boolean membership vector of $C_m(x_i)$ over the fixed catalogue. The representation contains no reference label, loss, or delivered-only filter. A row with one true entry is a precise delivery, a row with several true entries is an ambiguous or coarsened delivery, and an all-false row is non-delivery. Keeping the empty set inside the output type is important: it ensures that an abstained parcel remains in the population rather than disappearing before it can be priced.

The baseline emits the singleton containing the posterior maximizer. Both catalogue-shrinking arms apply the same operator to a different predeclared subcatalogue. If the unrestricted posterior maximizer belongs to that subcatalogue, it is emitted; otherwise the output is empty. The operator does not inspect the reference label. The two arms therefore differ only in the training-side rule that orders classes for withdrawal, not in how a test parcel is handled.

Per-parcel abstention emits the posterior maximizer when its confidence clears a fixed threshold and the empty set otherwise. Set-valued prediction sorts labels by posterior probability and emits the smallest non-empty prefix reaching a calibrated cumulative mass. The constructor applies that mass but never estimates it: calibration belongs entirely to training and validation, and any coverage guarantee depends on that preceding calibration rather than on the constructor alone.

Taxonomic back-off uses a predeclared partition of the leaf catalogue. A confident parcel keeps its singleton; an uncertain parcel receives every leaf in the coarse group containing the posterior maximizer. The implementation rejects a partition with missing, repeated, or out-of-catalogue leaves. Consequently, a malformed hierarchy cannot silently turn a parcel into non-delivery. The current comparator implements one named back-off level; a deeper hierarchy would be a different operating rule and is not selected after observing test outcomes.

All operating points—the retained catalogue, confidence threshold, calibrated mass, and coarse partition—are inputs to these operators. None is fitted by them, and none may be re-equalized on the held-out block. This separation is what makes test-label invariance inspectable: changing the test labels cannot change a delivered set.

Invariance to the held-out labels is a property of the operators; disjointness of the blocks is a property of whoever calls them, and the implementation no longer assumes it. A fitted operating point carries the stable parcel identities of the rows used to fit it and refuses to be applied to any of them. Identity is not inferred from posterior bytes: distinct parcels may have identical posteriors, and recomputing the same parcel may change its bytes. Handing the evaluated block back as calibration data therefore fails rather than passing quietly, which is how the corresponding leak entered an earlier version of this work.

The implementation makes this temporal order part of the interface. Its fitting function accepts training and validation arrays but has no held-out argument; its application function accepts held-out posteriors but has no label argument; only the evaluator receives labels, after all six membership matrices exist. A single orchestration path then composes these stages without supplying a default catalogue size, delivery target, miscoverage level, back-off rate, hierarchy, or loss table. This prevents an absent preregistration choice from becoming a library default.

The leaf catalogue is fixed by the training-side model output. Separately, a neutral evaluation universe is selected once from training labels using the predeclared support floor. Validation and held-out labels cannot add or remove a class from that universe. Every arm retains every held-out parcel in its loss denominator, including non-deliveries and parcels whose true class lies outside the neutral classwise summary. A selected class absent from a held-out block remains visible with zero support and an undefined classwise loss; it is not silently dropped or converted to a valid zero. Thus the primary parcel mean and every classwise diagnostic use denominators fixed before the arm's delivery is observed.

4.2 The Complete Action–Outcome Grid

The loss interface follows the full grid fixed by the elicitation instrument. A precise label is either correct, the zero-loss reference, or wrong. An empty set is non-response. A multi-label set either contains or excludes the truth. A coarse class likewise either contains or excludes the truth. These are seven cells in total. The two failure cells for non-singleton answers are indispensable: without them, a prediction set or a coarse answer would be charged only when it succeeds and would again acquire a free failure mode.

Classification into a cell follows the delivery visible to the user, not merely the name of the algorithm. A singleton emitted by the set-valued arm is scored as a precise label. A multi-label delivery is distinguished as an unconstrained label set or a named coarse class, because the two may impose different work on the affected party. An empty row is non-response under every arm. Every parcel maps to exactly one cell.

For an affected party a, the evaluator accepts a complete mapping from those cells to non-negative finite losses. It has no default values, requires the correct precise label to remain the zero reference, and refuses an incomplete mapping. The empirical values are deliberately not specified here: they are the output of the elicitation, not hyperparameters of the model. Synthetic tables are used only to test the mechanism, including the necessary property that changing the relative price of a wrong label and silence can reverse which action has lower loss.

Finally, the evaluator averages the resulting parcel losses over all eligible parcels in the held-out block, exactly as in Eq. (2). Non-deliveries therefore contribute their declared loss and their count to the same denominator as every other outcome. Expected set size is reported separately as a descriptor. The earlier size-based utility remains useful for a diagnostic link to macro recall, but it is not the common currency used to rank the six arms.

5 Evaluation Protocol

The protocol is executable up to the value of the affected-party loss table. Nothing in this section supplies that missing value, and no empirical contrast is run in order to write it.

5.1 Split Order and Pair Formation

For each dataset and eligible predictor, training fixes the neutral class universe and the support-based catalogue; validation fixes the remaining operating-point components; and the held-out block is touched only after those choices are immutable. Every parcel carries a stable identity across these stages. Training and validation identities must be disjoint, and applying a fitted operating point to any identity seen in either split is an error. This check uses parcel identity rather than posterior equality, which avoids both accepting a recomputed copy of a parcel and rejecting two distinct parcels whose posteriors happen to coincide.

The evaluator then records one loss for every eligible held-out parcel under every arm. Pairing is by parcel identity, not row position: one arm may be reordered, but it may not omit, duplicate, or substitute a parcel. Its spatial-block and `patch_id` metadata must agree with the comparator for that identity. The training-derived class universe travels with the loss vector and cannot be reconstructed from a held-out sample or a bootstrap draw.

5.2 Two Declared Uncertainty Views

The paired contrast is left-arm loss minus comparator loss, so a negative value favours the left arm. The primary, patch-cluster view is centred on the mean of the paired parcel losses in Eq. 2. It samples whole `patch_id` groups with replacement within each block, carries every parcel of a sampled patch together, and combines the resampled block means using the original parcel shares of those blocks. There is no parcel-level bootstrap.

Separately, parcel differences are averaged within each spatial block and those block means are given equal weight. A paired Student interval over them exposes between-block variation; it does not turn overlapping folds into independent regions. This is explicitly a sensitivity estimand: when block sizes differ, its centre need not equal the parcel-population contrast. Forcing both centres to coincide would silently replace Eq. 2 with an equal-block target.

Both views retain the complete parcel denominator and are reported side by side. If a view has fewer than three unique paired units, the implementation returns the observed block deltas and a reason, but no interval, no p-value, and no eligibility for a Holm family. A comparison declared identical by construction before evaluation is retained as a degenerate self-check and explicitly excluded from multiplicity rather than inflating the exploratory family.

5.3 Spatial Partition Selection and Sensitivity

The number of spatial blocks is selected before arm losses are evaluated. The rule sees only the exact minimum test-to-training centroid distance and the buffer used to construct the folds: the former must be at least one order of magnitude larger than the latter. This is a declared design threshold, not a proof of spatial independence; centroid distance is an upper bound on boundary separation, and residual spatial autocorrelation is not diagnosed here. Under the sealed profile, the rule uniquely selects five blocks. If no candidate or more than one candidate passes, the implementation stops instead of inventing a smallest- or largest-k tie-breaker.

An earlier rule also required a minimum number of supported classes in the worst block. That condition belonged to a retired classwise macro F-score estimand. It cannot identify the current mean loss over all eligible parcels and therefore remains a composition diagnostic rather than a selection condition. This distinction prevents a criterion made obsolete by the estimand from surviving merely because it still selects the same value.

Selection and sensitivity are separate interfaces. After the partition is fixed, every declared k is applied to the same paired parcel-loss vectors and the entire curve is reported. Partition fingerprints are invariant to row order and arbitrary block names, while duplicate k values, effective block counts that differ from the requested value, and adjacent parcel universes are errors. Finer partitions thus expose sensitivity to spatial grouping; they never enter the analysis as additional independent evidence or as a route to statistical power.

5.4 Power Has the Same Unit as Its Test

The minimum detectable effect for the paired block analysis is obtained from the noncentral Student distribution, not by adding two central-t quantiles. For

n independent paired blocks, degrees of freedom $\nu = n - 1$, assumed standard deviation σ of the paired block differences, and test-wise level α^*, let

$$c = t_{1-\alpha^*/2,\nu}, \qquad \lambda = \frac{\delta\sqrt{n}}{\sigma}. \tag{3}$$

The implementation solves for the smallest absolute δ such that

$$\Pr\{T_\nu(\lambda) < -c\} + \Pr\{T_\nu(\lambda) > c\} \geq 1 - \beta. \tag{4}$$

All inputs are explicit. Repeated unit identifiers cannot increase n, fewer than three units are not assigned a publishable MDE under our declared rule, and zero observed variance is not converted into a zero MDE. For an unadjusted comparison $\alpha^* = \alpha$. For a planned Holm family with m comparisons, prospective reporting uses the first-step threshold $\alpha^* = \alpha/m$; it is not described as one universal Holm level for every ordered test.

This calculation belongs only to the equal-block sensitivity estimand above. It does not describe the primary patch-cluster bootstrap, whose power depends on the elicited loss distribution and the observed cluster structure and must therefore be evaluated by simulation after that loss exists. Calling the block MDE the power of the primary estimand would recreate the unit mismatch that the two-view protocol was introduced to prevent. The code is executable on synthetic inputs, but no empirical MDE is reported here.

The primary comparison is reported without multiplicity adjustment and is never passed to the exploratory correction. Each exploratory Holm family contains intervals from one declared inferential unit only: block-based and patch-based p-values cannot be mixed. A structurally identical comparison is the sole declared exclusion. If any other planned comparison lacks a valid p-value—because there are too few units or because between-unit variance is exactly zero—the implementation does not reduce the family after seeing that failure; Holm is not applied to that family and the reason is reported.

Identity is a design claim, not a pattern inferred from the observed losses. Consequently, a zero contrast with zero resampling variance receives no p-value and cannot exclude itself from Holm; the caller must supply the identities declared before evaluation, and an unknown identity name is an error. A declared identity is accepted but not taken on trust: two arms identical by construction cannot differ on any parcel, so the implementation checks the paired parcel losses before aggregation and rejects a declaration contradicted by even one row. Checking only the interval status and its centre would leave a cancellation path open, especially below the minimum-unit floor where no interval exists. The asymmetry is deliberate—observed equality cannot establish an identity, which is why the caller declares it, while an observed difference refutes one—and it prevents a family from shrinking after its p-values are visible.

The confidence level, bootstrap count, random seed, and minimum-unit rule are required inputs with no defaults. Their eventual values belong to the signed preregistration. Until the affected-party loss table and those values exist, this section specifies behavior and failure modes only; it does not license an empirical result.

6 Results

7 Discussion

A Appendix

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